OFFICIAL PUBLIC SYSTEM GUIDE · 23 OCTOBER 2026

ADAM — Governed State & Transformations

Governed state changes, policy enforcement, traceability and replay

BTG-LIB-ADAM-SG · Edition 1.0 · Audience: Authorized operators, governance reviewers, rights holders and institutional stakeholders · Issued 23 October 2026

Publication boundary: This is an explanation of the system and its public operating model, not a source release, engineering specification, certification or promise of live service availability. Product status is stated with the limitations below.

Executive overview

ADAM is the authorized transformation and state-management layer in the BTG architecture. It is designed to apply permitted changes under recorded authority while preserving the ability to understand what happened and, where applicable, replay or verify a deterministic sequence. ADAM does not independently create ownership, payment or regulatory approval.

01 — Purpose and place in BTG

Modern digital systems change constantly: a research result is admitted, a right is granted or withdrawn, a record is updated, or a workflow moves between states. ADAM is intended to govern such changes under an explicit rule and identified authority.

ENTITY supplies the identity, rights and evidence context. BTDU retains authoritative knowledge; FarmNodes and NIKI generate candidate work; the Exchange concerns separately valid commercial and financial actions.

02 — What counts as a protected change

Some transitions are harmless UI changes; others alter access, release status, entitlement, financial records or evidence. A protected transition should require an authorized actor or system role, defined preconditions and a traceable outcome.

A tool having technical permission to write a database is not proof that the person operating it has legal authority to sell an asset or grant a third-party right.

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03 — Authority and decision boundaries

A rights holder or valid delegate may authorize a defined action in a specified scope. Policies may constrain recipient, purpose, jurisdiction, validity period and revocation conditions.

ADAM is expected to enforce known rules and preserve evidence of decisions. It cannot determine disputed legal title, invent missing consent or substitute for a regulator’s ruling.

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04 — Deterministic state and replay

Where an action is deterministic, replay can help verify that the same valid prior state and approved event sequence lead to the same resulting state under the same versioned rules.

Reconstruction of a machine state is not a reversal of an external-world contract, payment or personal-data disclosure. External acknowledgements must be separately reconciled.

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05 — Evidence and audit trail

An adequate action record should identify the relevant authority, action class, object, version, timing, conditions, outcome and any independent acknowledgement. The public audit view may mask private identities or sensitive payloads.

Hashes and signatures strengthen integrity checks, but do not prove that an input declaration was factually true or a person possessed legal authority in the first place.

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06 — Approval, denial and exception cases

A denied transition should leave a clear reason class without leaking confidential policy details. A disputed grant, expired authority or unresolved verification should stop the affected protected operation until reviewed.

An operator should not bypass denial by switching interfaces. Repair involves correcting the underlying legitimate authorization or data condition, not suppressing enforcement.

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07 — Connection to NIKI and FarmNodes

Research systems can propose new algorithms, Circuits or assets, but promotion to a trusted registry or commercial workflow requires independent evidence and the correct approval. ADAM is intended to enforce the gate between candidate and accepted state.

NIKI cannot sign for an owner or grant itself unrestricted access. A FarmNode’s internal success result does not override the external rights or qualification rules.

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08 — Relationship to Bridge and the Exchange

Bridge permission changes and governed data releases are examples of actions requiring a precise audit trail. Exchange instrument states, rights changes and settlement evidence need distinct legally accountable decision-makers.

A blockchain-like or internal ledger record does not alone complete fiat settlement or legally transfer a security. ADAM is a governed execution layer, not a licensed market operator.

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09 — Recovery and contested records

After outages or inconsistent messages, operators need to determine the last verified state, pending actions, authoritative records and safe reconciliation route. Replay should expose mismatches instead of silently fabricating success.

A dispute may require correction records, rights suspension or external mediation. The historical event record should remain intelligible even when a later authoritative decision supersedes it.

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10 — Current qualification and disclosure boundary

Local host tests and deterministic transformation demonstrations may show important components of this architecture. They do not independently establish owner-signed production authority, multi-host resilience or third-party recognition of external claims.

Public pages explain roles, permissions and evidence expectations. They intentionally exclude production authorization rules, secrets, internal security mechanisms and private source.

Related manuals and use

The companion User & Operator Manual describes the decisions and reviews applicable to this system. References to proposed user flows are explicitly conditional on qualification of the relevant product release.

Updated 23 October 2026 · Editorial documentation, not proof of regulatory authorization or release readiness · Proprietary implementation withheld

Expanded system explanation

Question	Clear answer
Audience and purpose	ADAM — Governed State & Transformations is described for the public, institutional evaluators and prospective partners; this is not a software implementation reference.
What exists	A governance and state-change layer intended to allow only authorized transitions in the BTG system while recording reasons and evidence.
Analogy	A controlled transaction log and gatekeeper: it checks whether a particular change is permitted, records the decision, and supports reviewing what happened.
Authority	The described process requires actual contracts, human authority and technical qualification at every relevant boundary.

Practical examples — what a reader should infer

Permission expiry

An approved Bridge research permission expires. A later attempted transfer is rejected and logged with a reason code, rather than allowed by the existence of an old token.

Safe promotion

A test passed for a candidate Circuit but production owner authority is absent. ADAM can keep the protected promotion on hold.

Audit replay

A reviewer recreates a known sequence using the same approved inputs and versions to check the recorded state history, without assuming replay can alter external bank balances.

System lifecycle — each separate decision

1. Receive a proposed protected change and its target
2. Identify requester, source of authority, scope and expiry
3. Evaluate current state and applicable policy
4. Approve, deny or defer without leaking sensitive rules
5. Record signed or verifiable decision context
6. Verify resulting state and handle exceptions or replay within approved scope

People, organizations and responsible roles

- Requester proposes a defined action
- ENTITY may provide identity/rights references
- ADAM applies the configured authorization and state rules
- Owner or independent reviewer handles exceptions, appeals and promotion

Where the system can fail

- A local cryptographic signature is not evidence of legal authority by itself.
- Incorrect policies can allow or block legitimate work.
- Replay cannot undo outside-world events or lawfully erase financial/contractual records.
- Production owner identity, signed grants and revocation must be demonstrated with actual enrolled targets.

Evidence a serious evaluator would require

These are acceptance requirements, not claims that each artifact has been issued or independently certified:

1. Production owner authority chain and grant revocation
2. State transition receipts with time and actor
3. Idempotency / duplicate side-effect safeguards
4. Negative and conflict test results
5. Cross-host evaluation when distributed
6. Governance change-control and escalation rules

Terms explained in ordinary language

Term	Meaning
State transition	A recorded move of a governed object from one permitted condition to another.
Grant	Scoped authority enabling a particular action.
Replay	Re-running a deterministic record for validation where permitted.

Term	Meaning
Idempotence	Protection against unintended duplicate effects of repeated operations.

How to read the current status accurately

The architecture and local engineering records are distinct from production operation. A named component or test figure cannot establish authorized commercial rights, fully qualified infrastructure or independently accepted financial activity. Read the dated corporate evidence register for each claimed version; ask for a scoped, access-controlled evidence bundle for decisions that rely on private implementation.

More detailed reading: User & Operator Manual · Non-source engineering evaluation · Regulatory assessment pathway · Complete cross-system example.